

REPUBLIC OF ZAMBIA

MINISTRY OF EDUCATION

iLearnZed Study Series

CHEMISTRY

Complete Form 1 Master Notes

EXPLORING MOLECULES & REACTIONS

Secondary Chemistry for Zambia

GRADE 10



Developed strictly in alignment with the approved 2023 Curriculum Framework.
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CURRICULUM DEVELOPMENT CENTRE (CDC) COMPLIANT

iLearnZed Platform Curriculum Series

Form 1 Chemistry Complete Study Notes

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Foreword & Preface

Welcome to the **iLearnZed Master Notes Series**, a world-class educational resource engineered to connect curriculum alignment with active learning outcomes. Our core philosophy is to provide rigorous, clear, and highly localized study materials that motivate learners and enhance cognitive mastery.

These complete Form 1 Chemistry Study Notes are developed in strict compliance with the ****2023 Revised** **Zambian Curriculum Framework**** (CDC). They bridge abstract chemical concepts with the everyday realities of Zambia's geography, agriculture, and mining industries.

By partnering with **Primetutor Education Limited**, based in the historic mining town of Luanshya, iLearnZed is committed to bringing state-of-the-art digital and print textbook resources to schools, colleges, and self-directed learners across sub-Saharan Africa. This series is designed not only to prepare you to pass your Examinations Council of Zambia (ECZ) national exams but to inspire a lifelong passion for scientific inquiry and sustainable development.

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Luanshya, Copperbelt Province, Zambia

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Editorial Committee

iLearnZed Learning Systems

Lusaka/Luanshya, Zambia

Chapter 1: Introduction to Chemistry

Learning Objectives:

- Define Chemistry and identify its five main branches.
- Evaluate the impact of chemistry on everyday life, agriculture, medicine, and mining in Zambia.
- Apply standard laboratory safety precautions, first aid, and waste management.
- Identify and manipulate core chemical apparatus precisely.

1.1 What is Chemistry & Its Branches

Every time you strike a match in your kitchen, boil water for breakfast, or wash your hands with soap, you are conducting chemical experiments!

Chemistry is defined as the scientific study of the composition, structure, properties, and changes of matter. Matter is anything that possesses mass and occupies physical space. Because all physical objects are made of matter, chemistry is the "Central Science" that links physics, biology, and environmental sciences.

The Five Main Branches of Chemistry:

- Inorganic Chemistry:** Focuses on inorganic substances (compounds not containing carbon-hydrogen bonds) such as minerals, ores, metals, and salts. This branch is directly tied to the mineral processing industry in Luanshya and the Copperbelt.
- Organic Chemistry:** The study of carbon compounds (excluding carbonates and simple oxides). It covers fuels, plastics, polymers, and synthetic compounds.
- Physical Chemistry:** Explores the physical principles governing chemical reactions, energy changes (thermodynamics), and chemical kinetics.
- Analytical Chemistry:** Relies on qualitative and quantitative analysis of substances (e.g., determining the concentration of lead in soil samples near local mine shafts).
- Biochemistry:** The study of chemical reactions occurring in living organisms, such as cellular respiration and plant photosynthesis.

1.2 Importance of Chemistry & Local Industry

Zambia's economy and daily life rely heavily on chemical engineering and processing:

- **Agriculture & Soil Health:** Production of synthetic fertilizers (such as Ammonium Nitrate and Compound D) at the Nitrogen Chemicals of Zambia (NCZ) factory in Kafue. Chemistry allows farmers in Central Province to neutralize acidic soils using calcium carbonate (lime).
- **Mining & Minerals:** Copper smelting, froth flotation, and electrolytic refining at the Luanshya, Mufulira, and Nchanga mines. High-grade copper (99.99% pure) is produced using electrical chemistry.
- **Medicine & Healthcare:** Formulating pharmaceutical medicines, intravenous fluids, and sterilizing agents to protect public health.
- **Manufacturing:** Crafting detergents, vegetable oils, plastics, and building materials (cement from Chilanga Cement plants).

1.3 Laboratory Safety & Waste Management

A science laboratory is an exciting workspace, but it can be dangerous due to toxic chemicals, corrosive acids, flammable gases, and fragile glassware. Adhering to safety protocols is a mandatory competence.

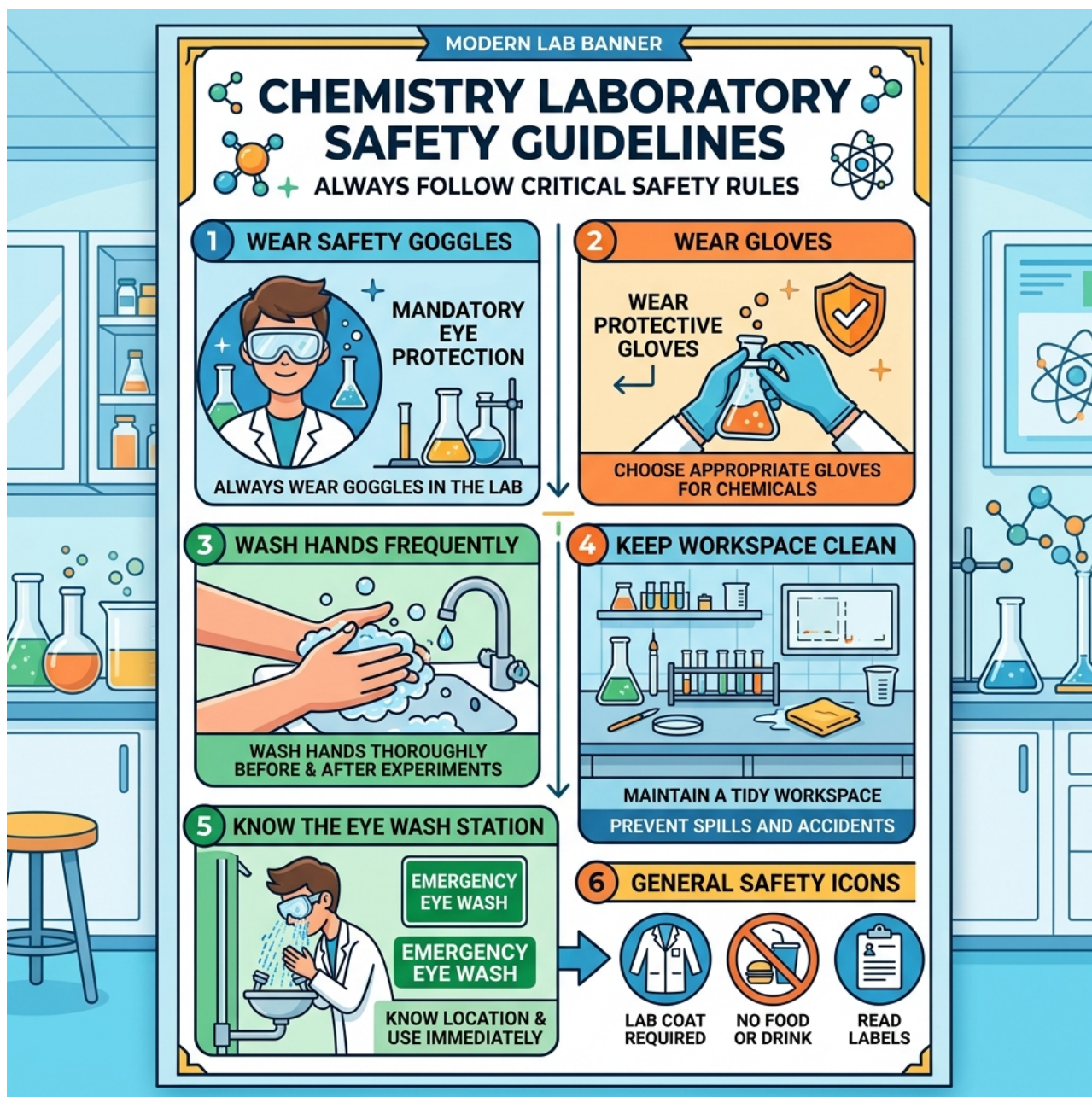


Figure 1.1: Core Chemistry Laboratory Safety Practices and Hazards Guide.

Laboratory Rules & Precautions:

- Always wear personal protective equipment (PPE): safety goggles, chemical-resistant gloves, a buttoned lab coat, and closed leather shoes.
- Never smell, touch, or taste chemicals directly. To smell a gas, gently wave (waft) the vapor towards your nose.
- Never run, eat, or drink in the laboratory.
- In case of a chemical burn, wash the affected area under clean, running tap water for at least 15 minutes and notify the instructor immediately.

Solid and Liquid Waste Disposal Protocol:

Disposing of laboratory waste incorrectly leads to toxic pollution and damages plumbing:

- **Acids and Alkalis:** Must be carefully neutralized (e.g., mixing weak acid/base) before being flushed with a large volume of water. Concentrated acids must be poured into a specific toxic waste container.
- **Broken Glass:** Must be cleaned up immediately using a brush and dustpan and placed in a dedicated "Broken Glassware Box" to prevent puncture wounds.
- **Solid Residues:** Used matches, charcoal pieces, and soiled filter papers must be discarded in solid trash bins, never in sinks.

1.4 Apparatus and Precision

Accuracy in Chemistry can be the difference between a successful synthesis and a dangerous explosion. You must know which apparatus is appropriate for different tasks:

Apparatus Image / Name	Scientific Function	Precision Limit
Beaker / Conical Flask	Approximate mixing, heating, and holding of liquid reagents. They are never used for volumetric measurements.	Very Low ($\pm 10\%$ error margin)
Measuring Cylinder	Measuring variable liquid volumes quickly (e.g., 50 cubic cm).	Medium ($\pm 0.5 \text{ cm}^3$)
Burette	Delivering precise, variable volumes during quantitative experiments like acid-base titrations.	High ($\pm 0.05 \text{ cm}^3$)
Pipette	Delivering a single, extremely precise fixed volume of liquid (usually 25.0 cm^3 or 10.0 cm^3).	Very High ($\pm 0.02 \text{ cm}^3$)
Triple-Beam Balance	Determining the physical mass of solid substances precisely.	High ($\pm 0.01 \text{ g}$)

Topic Checkpoint:

1. Identify the main branch of chemistry involved in calculating the reaction speed (rate of reaction) at different temperatures.
2. Explain the proper step-by-step procedure to manage an accidental acid spill on a student's hand during a practical session.
3. A student wants to measure exactly 24.3 cm^3 of hydrochloric acid. Which volumetric apparatus should they select?

Chapter 2: Matter & Phase Changes

Learning Objectives:

Define matter and distinguish its units (atoms, molecules, and ions).

Differentiate the arrangement, motion, and attractive forces in solid, liquid, and gas states.

Explain diffusion and Brownian motion as empirical proof of the Kinetic Theory.

Construct, label, and interpret heating and cooling curves of pure substances.

2.1 The States and Basic Units of Matter

Matter is anything that occupies space and possesses mass. It is composed of tiny, discrete particles. The fundamental units of matter are:

- **Atoms:** The smallest neutral particle of a chemical element that retains its chemical properties (e.g., Helium, Iron).
- **Molecules:** A group of two or more atoms chemically bonded together (e.g., Oxygen gas, O_2 , or Water, H_2O).
- **Ions:** Charged atoms or groups of atoms formed when neutral atoms lose or gain electrons (e.g., Copper ion, Cu^{2+} , or Sulfate ion, SO_4^{2-}).

Matter exists in three primary physical states: **Solid, Liquid, and Gas**. Under extremely high energy conditions, matter transitions into **Plasma** (ionized gas where electrons are stripped from nuclei, found in stars and lightning).

Particulate Arrangement in States of Matter

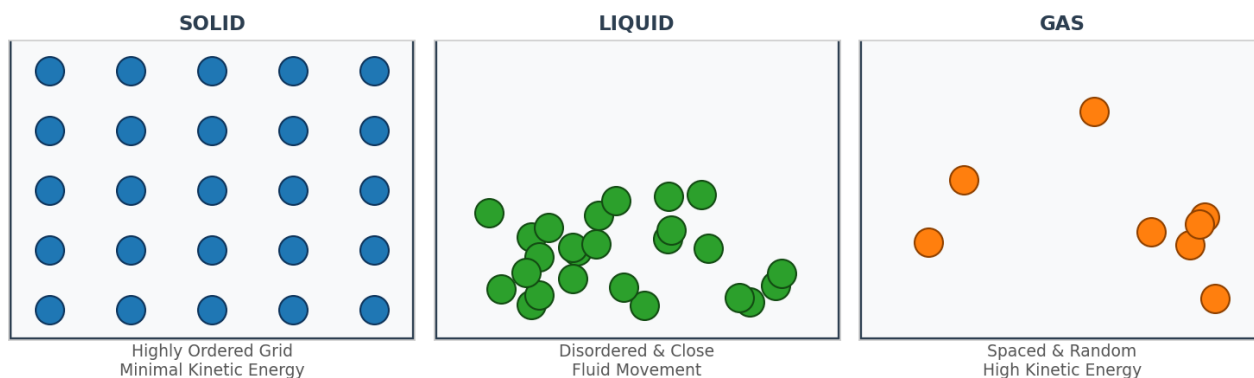


Figure 2.1: Particulate arrangement models in Solid, Liquid, and Gas containers.

2.2 Particle Arrangements & Kinetic Theory

According to the **Kinetic Theory of Matter**, all matter consists of particles that are in continuous, random motion. The average kinetic energy of these particles is directly proportional to the temperature in Kelvin.

Comparative Analysis of States:

- Solids (Ice, Copper):** Particles are tightly packed in a highly ordered three-dimensional lattice. Strong electrostatic attractive forces keep them in fixed positions. They cannot move freely; they only vibrate. Solids have a fixed shape and volume and are incompressible.
- Liquids (Water, Cooking Oil):** Particles are close together but arranged randomly. The attractive forces are weaker than in solids, allowing particles to slide over and past each other. Liquids have a fixed volume but take the shape of their container. They are virtually incompressible.
- Gases (Oxygen, Steam):** Particles are spaced extremely far apart. Intermolecular forces are negligible. They move randomly at very high speeds in straight lines, colliding with each other and the container walls. Gases have no fixed shape or volume and are highly compressible.

Scientific Proof of Kinetic Theory

We cannot see gas or liquid molecules with the naked eye, but we can verify their motion:

- 1. Diffusion:** The movement of particles from a region of higher concentration to a region of lower concentration, down a concentration gradient. If a drop of purple potassium manganate(VII) crystals is dropped into a beaker of water, the color spreads through the water without stirring.
- 2. Brownian Motion:** The continuous, erratic, zig-zag motion of suspended particles (e.g., smoke particles in a smoke cell) caused by the random bombardment of invisible air molecules.

2.3 Phase Changes & Heating Curves

Adding or removing thermal energy (heat) alters the kinetic energy of particles, causing a change in physical state.

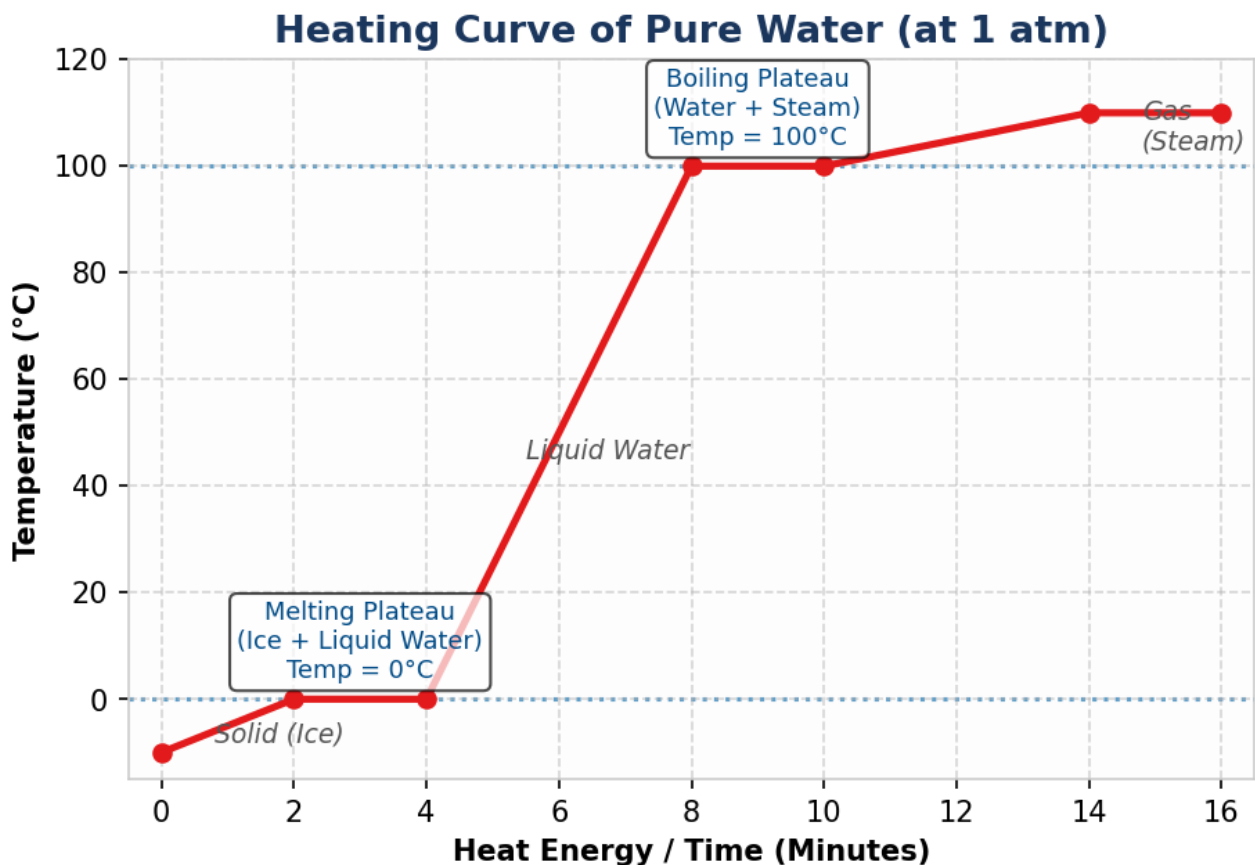


Figure 2.2: Temperature-time graph representing the heating curve of pure water.

Deconstructing the Heating Curve (Figure 2.2):

- **Solid phase heating:** Temperature rises as ice absorbs thermal energy, increasing particle vibration.
- **Melting Plateau (0°C):** The temperature remains constant. The heat energy absorbed is used to break the strong bonds holding the ice lattice together rather than raising the temperature. This is the melting point.
- **Liquid phase heating:** Liquid water absorbs heat, increasing particle kinetic energy and temperature.
- **Boiling Plateau (100°C):** The temperature flattens. The heat energy is absorbed to overcome all remaining intermolecular forces, converting liquid molecules into gaseous steam.

Topic Checkpoint:

1. Differentiate between an atom, a molecule, and an ion.
2. Using the Kinetic Theory, explain why gases diffuse much faster than liquids.
3. Define the term *melting point* and explain why a flat line (plateau) appears on a heating curve during melting.

Chapter 3: Elements, Compounds & Mixtures

Learning Objectives:

Classify substances into elements, compounds, and mixtures.

Distinguish physical changes from permanent chemical changes.

Analyze melting/boiling points and chromatography to assess purity.

Demonstrate proficiency in setting up and selecting filtration, distillation, and chromatography operations.

3.1 Pure and Impure Matter

Matter can be classified based on its chemical purity:

- Element:** A pure substance composed of only one type of atom (e.g., Oxygen, 8^{16}O , and Iron, 26^{56}Fe). Elements cannot be split into simpler substances by chemical methods.
- Compound:** A pure substance composed of two or more elements chemically combined in fixed ratios (e.g., Carbon Dioxide, CO_2 , and Copper(II) Sulfate, CuSO_4). The properties of a compound differ completely from its constituent elements.
- Mixture:** An impure substance consisting of two or more substances physically mixed in any proportion (e.g., sandy water, mineral ores). The components retain their original properties and can be separated easily by physical methods.

3.2 Physical vs. Chemical Changes

Physical changes alter only state or shape, whereas chemical changes form entirely new molecules:

Physical Change Characteristics	Chemical Change Characteristics
No new chemical substance is formed.	One or more new chemical substances are created.
Highly reversible by physical cooling/heating.	Irreversible by simple physical means.

No change in the mass of the substances involved.	Mass change occurs as reactants form products.
Minimal thermal energy is released or absorbed.	Large energy changes occur (exothermic or endothermic).
Example: Sublimation of iodine, freezing water.	Example: Rusting of iron, burning magnesium.

3.3 Criteria of Purity

In pharmaceutical and domestic sectors, knowing the purity of a compound is critical. Pure substances have sharp, specific melting and boiling points:

- **Melting & Boiling Points:** A pure substance melts and boils at a precise temperature (e.g., pure water boils at exactly 100°C). Impurities **lower the melting point** and **elevate the boiling point**, causing phase transitions to occur over a broad temperature range.
- **Chromatography:** Pure substances produce a **single spot** on a chromatogram, whereas mixtures separate into multiple spots.

3.4 Industrial Separation & Purification

Chemists use physical techniques to isolate pure elements or compounds:

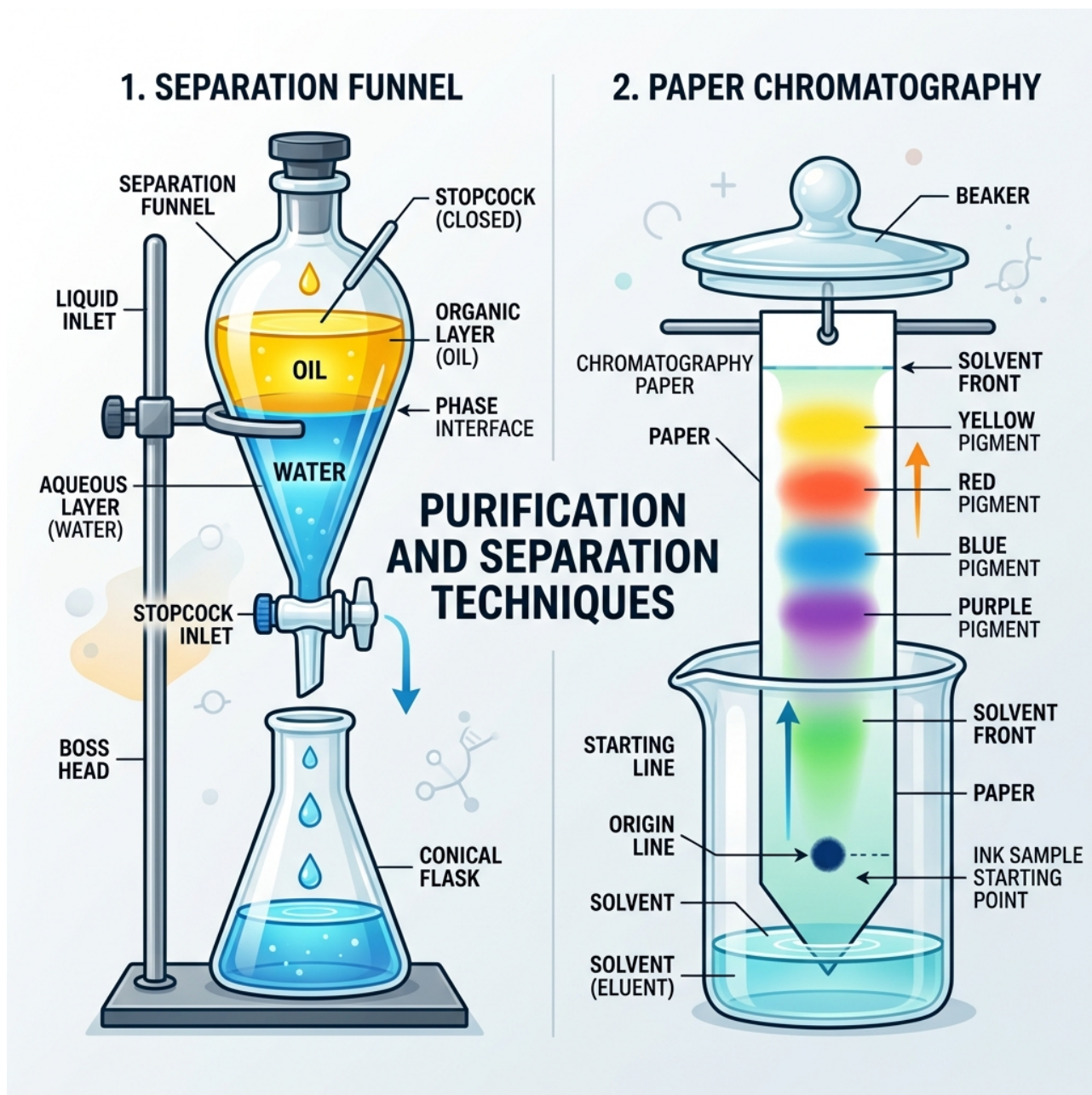


Figure 3.1: Technical setups for separating funnel extraction and paper chromatography.

Key Separation Methods:

1. **Filtration:** Separates an insoluble solid from a liquid. The solid caught on the filter paper is the **residue**, and the clear liquid is the **filtrate**.
2. **Crystallization:** Separates a soluble salt from its saturated solution by evaporating excess solvent, leaving pure crystal lattices.
3. **Simple Distillation:** Separates a volatile liquid solvent from a non-volatile solid solute (e.g., purifying water from saline solutions).
4. **Fractional Distillation:** Separates miscible liquids with close boiling points (e.g., separating ethanol, B.P. 78°C, and water, B.P. 100°C) using a fractionating column packed with glass beads.
5. **Paper Chromatography:** Separates different solutes based on their relative solubility in a mobile phase (solvent) and adsorption on a stationary phase (filter paper), as shown in Figure 3.1.

iLearnZed Lab Activity: Paper Chromatography

Objective: Separate components of black dye and calculate Retention Factor (R_f) values.

Apparatus: Chromatography paper, solvent (ethanol/water), pencil, ruler, capillary tube, beaker, and black ink.

Procedure:

1. Draw a pencil line 2 cm from the bottom of the chromatography paper (the origin line).
2. Spot the black ink on the origin line and let it dry.
3. Suspend the paper in a beaker with 1 cm of solvent, ensuring the ink spot is **above** the solvent level.
4. Cover the beaker to prevent solvent evaporation. Allow the solvent front to rise to 8 cm.
5. Remove the paper, mark the solvent front, and measure the distance traveled by each dye.

Calculation: $R_f = \frac{\text{Distance traveled by solute spot}}{\text{Distance traveled by solvent front}}$

Chapter 4: Atomic Structure & Isotopes

Learning Objectives:

- Describe the structure of an atom and the properties of protons, neutrons, and electrons.
- Formulate expressions relating Atomic Number (Z), Mass Number (A), and Neutrons (N).
- Draw Bohr orbital electron shell configurations for the first 20 elements.
- Explain isotopes and calculate relative atomic mass from isotopic abundances.
- Evaluate the hazardous nature and medical/industrial benefits of radioactive isotopes.

4.1 Atomic Theory & Sub-atomic Particles

An **atom** is the smallest neutral particle of an element that can take part in a chemical reaction. According to the Bohr model, an atom consists of a central **nucleus** surrounded by concentric orbits or **electron shells**.

Fundamental Properties of Sub-atomic Particles:

Particle Name	Symbol	Relative Charge	Relative Mass (amu)	Location in Atom
Proton	p	+1 (Positive)	1	Inside the Nucleus
Neutron	n	0 (Neutral)	1	Inside the Nucleus
Electron	e ⁻	-1 (Negative)	1/1840	Outside (In Shells)

Atomic Neutrality: Atoms are electrically neutral because the number of positive protons inside the nucleus is exactly equal to the number of negative electrons in the electron shells.

4.2 Atomic Number, Mass Number & Nuclide Notation

Every element on the Periodic Table has a unique atomic identity:

- **Atomic Number (Z):** The number of protons in the nucleus of an atom. In a neutral atom, it is also equal to the number of electrons.
- **Mass Number (A):** The sum of protons and neutrons in the nucleus of an atom.

The number of neutrons (N) is calculated using the formula:

$$A = Z + N \text{ implies } N = A - Z$$

Nuclide Notation:

The nuclear composition of an atom is represented as:

$${}_Z^A \text{X}$$

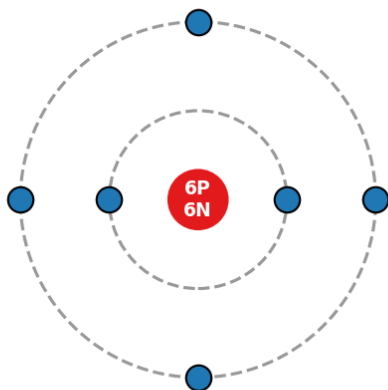
Where X is the chemical symbol of the element, A is the Mass Number, and Z is the Atomic Number.

4.3 Electronic Configurations (First 20 Elements)

Electrons occupy fixed shells at different energy levels. The maximum capacity of each shell is governed by the formula: $2n^2$ (where n is the shell number):

- **Shell 1 (Inner):** Holds a maximum of 2 electrons (the *Duplet* rule).
- **Shell 2:** Holds a maximum of 8 electrons (the *Octet* rule).
- **Shell 3:** Holds a maximum of 8 electrons for stable main-group elements.

Carbon Atom (C)
Atomic Number $Z=6$, Config: 2,4



Sodium Atom (Na)
Atomic Number $Z=11$, Config: 2,8,1

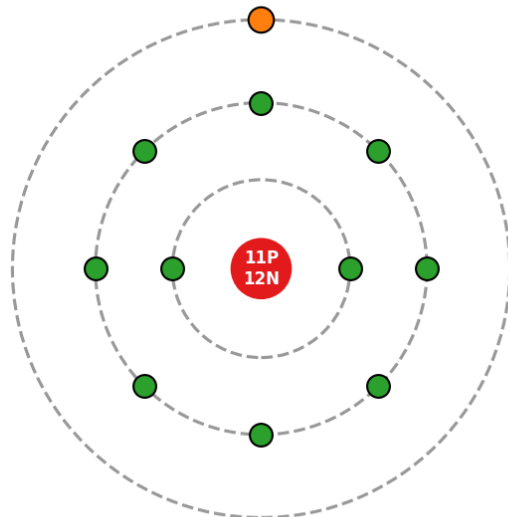


Figure 4.1: Bohr schematic electron shell structures for Carbon (2,4) and Sodium (2,8,1).

Carbon ($Z=6$) has 2 inner electrons and 4 outer valence electrons. Its electronic configuration is written as **2,4**. Sodium ($Z=11$) has a configuration of **2,8,1**, as illustrated in Figure 4.1.

4.4 Isotopes & Relative Atomic Mass

Isotopes are atoms of the same element with the same number of protons (same Atomic Number) but different numbers of neutrons (different Mass Numbers).

Relative Atomic Mass (A_r) Calculation:

The Relative Atomic Mass is the weighted average mass of the naturally occurring isotopes of an element relative to Carbon-12.

Example: Carbon exists as 98.9% Carbon-12 (^{12}C) and 1.1% Carbon-13 (^{13}C).

$$A_r = \left(\frac{98.9}{100} \times 12 \right) + \left(\frac{1.1}{100} \times 13 \right) = 11.868 + 0.143 = 12.011$$

Radioactive Isotopes (Radioisotopes):

Unstable isotopes undergo nuclear decay, emitting alpha (α), beta (β), or gamma (γ) radiation:

Beneficial Uses:

- ***Medicine:** Cobalt-60 (^{60}Co) is used in cancer radiotherapy to destroy tumors at the Cancer Diseases Hospital in Lusaka.
- ***Industry & Energy:** Uranium-235 (^{235}U) undergoes nuclear fission to generate electrical energy.
- ***Agriculture:** Carbon-14 (^{14}C) tracks carbon pathways in photosynthesis.
- **Hazards:** Uncontrolled radiation damages cells and causes DNA mutations and cancer. Radioactive waste remains hazardous for thousands of years, requiring storage in specialized lead containers.

Topic Checkpoint:

1. An atom is represented nuclidely as $^{39}_{19}\text{K}$. Calculate its protons, electrons, and neutrons, and write its electronic configuration.
2. Describe the structural similarity and difference between Carbon-12 and Carbon-14 isotopes.
3. Name one radioisotope used in agricultural tracer studies and describe its safety risks.

iLearnZed Summative Examination

EXAMINATIONS COUNCIL OF ZAMBIA (ECZ) FORMAT • CHEMISTRY FORM 1

Time: 2 Hours • Total Marks: 50

Instructions: Answer all questions in Section A and B. Answer only one question in Section C.

SECTION A: Multiple Choice Questions (10 Marks)

Choose the correct letter representing the best answer (A, B, C, or D).

- Which branch of chemistry is concerned with studying the industrial production of fertilizers and acids?
A) Physical Chemistry
B) Inorganic Chemistry
C) Biochemistry
D) Analytical Chemistry
- A student needs to measure exactly 25.0 cm³ of dilute hydrochloric acid. Which apparatus is the most suitable?
A) Beaker
B) Pipette
C) Conical Flask
D) Measuring Cylinder
- Which of the following describes the arrangement and motion of particles in liquid water?
A) Tightly packed in a lattice, only vibrating.
B) Spaced far apart, moving randomly at high speeds.
C) Close together but disordered, sliding over each other.
D) Completely static, no kinetic energy.
- During the boiling of pure water, the temperature remains constant at 100°C. This plateau occurs because heat energy is used to:
A) Create new covalent chemical bonds.
B) Overcome intermolecular forces to convert liquid into gas.
C) Increase the chemical kinetic energy of protons.
D) Decompose water into hydrogen and oxygen gases.

5. Which of the following is a pure compound?

- A) Salt Water
- B) Copper Metal
- C) Nitrogen Gas
- D) Carbon Dioxide

SECTION B: Structured Questions (25 Marks)

Question 1: A mixture of sand, salt, and water is brought to a school laboratory in Chipata.

- (a) State the separation technique you would use to obtain dry sand from the mixture. **[2 Marks]**
- (b) Draw a labeled diagram of the filtration setup, highlighting the residue and filtrate. **[4 Marks]**
- (c) Describe how you would obtain pure, dry salt crystals from the salt water filtrate. **[3 Marks]**

Question 2: An atom of Sodium is represented nuclidely as ${}_{11}^{23}\text{Na}$.

- (a) State the Atomic Number and Mass Number of this Sodium atom. **[2 Marks]**
- (b) Calculate the number of protons, neutrons, and electrons present. **[3 Marks]**
- (c) State the electronic configuration and draw the Bohr electron shells. **[3 Marks]**

SECTION C: School-Based Assessment (SBA) Project Guidelines (15 Marks)

Compulsory SBA Project: Purifying Contaminated Water in Our Community

Scenario: A community water source in a rural Zambian district has become contaminated with suspended mud, clay, and dissolved salts. Maize fields are suffering due to soil degradation.

Tasks:

1. Working in groups of 4, design and construct a low-resource filtration and solar distillation device using local, recycled plastic bottles, gravel, sand, and charcoal.
2. Test the purity of your filtered and distilled water using criteria of purity (boiling point).
3. Compile a 3-page scientific report containing your hypothesis, practical diagrams, data tables, and a recommendation for your local community.